

Report coding challenge

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This project tackles a familiar challenge in the banking world: deciding who should get a loan. With every approval comes a risk, and the goal is to make those decisions as wisely as possible. The core objective of this project is to develop an automated, accurate, and interpretable credit risk assessment tool for Prometis Bank. We aim to identify patterns associated with default risk and support more informed lending decisions. By using historical data from past applications, we've built a model that helps predict which clients are more likely to repay their loans—and which ones might not. The idea is to support the bank's decision-making, reduce unnecessary risk, and ensure compliance with emerging regulatory frameworks such as the EU AI Act.

Context

The primary goal is to approve loans that are likely to be repaid, thereby generating consistent profits and maintaining the financial stability and reputation of Prometis Bank. With the recommended predictive model, we aim not only to avoid approving loans to high-risk clients, but also to create a reliable stream of income from timely, low-risk borrowers.

Given the high-risk nature of loan approval and the compliance requirements of the EU AI Act, the model must adhere to strict standards of Data Governance. This includes transparency in data handling, and clear integration with the bank's corporate risk management framework. The data set was divided into 80% training and 20% test subsets to reduce overfitting and to allow reliable performance evaluation as new applications are received.

A comprehensive data preprocessing pipeline was implemented to ensure that the data set is free of errors or inconsistencies that might compromise model outcomes or increase institutional risk. This process strengthens the bank's data governance by ensuring protection, accuracy, and consistency in client information. Since data scaling was applied to the numeric inputs, it standardizes inputs, which enhances integration, improves model performance, and reduces operational costs while maximizing data value and minimizing risks.

Data Analysis

The dataset consists of 50,000 observations, which provides a solid foundation for training a predictive model. However, one important limitation must be acknowledged. The dataset includes only approved applications. Although this is relevant to the main objective, predicting repayment behavior, it introduces a selection bias. Including rejected applications in future analyses could improve model robustness by identifying features that correlate strongly with risk and provide additional insight into the reasons for application denial.

The features provided are diverse in nature, covering demographic, financial, and application-related aspects. There are 18 primary features to control and input in the predictive model, each contributing a distinct layer of information. To harmonize the data for modeling, we modified and adapted variable types when necessary—for example, recasting zip codes as ordered categories and encoding categorical variables with clear baselines to preserve interpretability. We also explored creating new features that interact across categories, such as debt-to-income and payment-to-income ratios, which uncovered latent financial pressure patterns. These engineered variables significantly enhanced predictive power.

After cleaning and encoding, we worked with 45 features in total. Categorical variables like education level, home ownership, and application type were one-hot encoded, while numeric variables were standardized to account for differences in scale—an important step, particularly for linear models. Moreover, we found no missing values in the dataset. Several numeric features, including income, loan amount, and savings, exhibited positive skewness. Exploratory analysis revealed clear differences between repaid and defaulted loans: defaulters tended to have lower income and credit scores, shorter employment histories, and higher debt-to-income ratios. Savings amounts showed a notable drop among high-risk clients. These patterns were consistent across both visual inspection and summary statistics.

	Model	Accuracy	Precision	Recall	F1 Score	AUC
	Logistic Regression	0.769	0.726	0.623	0.670	0.838
h	Random Forest	0.768	0.726	0.617	0.666	0.831
	Gradient Boosting	0.769	0.722	0.642	0.670	0.836
	Decision Tree	0.760	0.700	0.629	0.663	0.821

Table 1: Model performance metrics on the test set

Outlier analysis using the IQR method revealed that approximately 44% of the rows contained at least one outlier—mostly in income, loan amount, or savings. However, these values appear plausible and were retained. Correlation analysis revealed strong multicollinearity among monthly payment, loan-to-income ratio, and payment-to-income ratio. These were used carefully in model selection to avoid redundancy.

The target variable is binary and shows a class distribution of roughly 62% repaid and 38% defaulted. This mild imbalance was addressed by evaluating models on metrics like AUC and F1 score, which account for class imbalance and prioritize discriminatory power. Overall, the dataset required thoughtful preprocessing, but offered rich and interpretable signals for modeling risk.

One limitation in the dataset is the absence of external credit scores or prior delinquency history, which are typically strong predictors of loan behavior. Including these features in future iterations could further boost predictive performance. Additionally, behavioral signals such as late payment streaks or missed installments—common in financial institutions—were not available and could represent valuable future inputs.

Modeling

A significant portion of a bank’s profits comes from loans. However, if these loans are not repaid, they pose a direct threat to the bank’s financial stability. Therefore, the predictive system must demonstrate high accuracy to minimize the risk of approving high-risk applicants, which could lead to profit losses or broader financial consequences¹.

A well-functioning predictive system offers several advantages such as reducing human error and bias, and thus improving consistency in decision-making. Moreover, it saves time and resources, allowing personnel to focus on more complex or exceptional cases. Lastly, it lowers operational costs by streamlining the application review process.

To ensure its effectiveness, the model must undergo thorough testing using diverse validation techniques. This will help build confidence in its reliability and ensure it provides a sound basis for making loan approval decisions. To assess default risk, we tested four predictive models with increasing complexity and interpretability trade-offs: Logistic Regression, Random Forest, Gradient Boosting, and Decision Tree. Each model was selected for its relevance in credit risk modeling and trained on an 80/20 train-test split with stratified sampling to preserve class balance.

We began with a regularized Logistic Regression model as a baseline, offering strong interpretation and robustness. Hyperparameters (penalty type and regularization strength) were optimized using grid search and 5-fold cross-validation based on AUC. After identifying the top contributing features, we retrained a simpler model with only the 10 most influential variables, retaining most of the predictive power while enhancing transparency.

Next, we implemented Random Forest and Gradient Boosting classifiers, which are ensemble methods capable of capturing complex, non-linear interactions. Random Forest was tuned for tree depth and sample splitting criteria, while Gradient Boosting was further optimized for learning rate and regularization. Both models improved recall and AUC scores compared to Logistic Regression, though at the cost of some interpretability. We also included a single Decision Tree model to serve as a fully transparent benchmark. Despite its lower predictive power, it provides a straightforward visual logic for approvals and serves as a baseline for explainability.

All models were evaluated on metrics that capture both performance and fairness in credit classification: Accuracy, Precision, Recall, F1 Score, and AUC. Additionally, we generated ROC curves, lift curves, and feature importance plots to help understand better the results and the four models. The same test set was used for all models in order to compare them better. The table below summarizes each performance:

Gradient Boosting emerged as the top performer, with the highest recall and F1 score, suggesting its particular strength in identifying high-risk clients. Logistic Regression followed closely and provided superior interpretability—ideal for regulatory settings. Random Forest achieved similar results, though slightly underperformed on recall. The Decision Tree, while the most interpretable, demonstrated weaker overall performance. In summary, Gradient Boosting is recommended for deployment due to its strong performance, while Logistic Regression remains a valuable tool for transparency and stakeholder communication.

¹Historically, the failure to properly assess credit risk, such as during the 2008 financial crisis, has contributed to the formation of financial bubbles and widespread economic fallout.

All models showed AUC scores above 0.82, confirming strong discriminative ability. Importantly, these results were consistent across cross-validation folds, reinforcing model stability and generalizability. These results could have been validated through 5-fold cross-validation, as we try to do at the end of the code: models displayed stable performance, with mean AUC scores above 0.83 and standard deviations below 0.004, confirming generalizability.

Beyond aggregate metrics, we evaluated model performance across income quartiles, education levels, and loan purposes. Gradient Boosting consistently maintained higher recall across subgroups, particularly for lower-income applicants and debt consolidation loans. Additionally, lift curves demonstrated that the model is over $2.5\times$ more effective than random selection in identifying defaults within the top decile of risk scores. These results suggest not only overall strength but also subgroup consistency—critical for equitable deployment.

Compliance and Governance

To ensure ethical and reliable deployment, the model was developed with close attention to the requirements of the EU AI Act and broader principles of responsible AI. From data preprocessing to model selection and evaluation, we prioritized transparency, fairness, and risk mitigation. A significant portion of a bank's profits comes from loans. However, if these loans are not repaid, they pose a direct threat to the bank's financial stability. Therefore, the predictive system must demonstrate high accuracy to minimize the risk of approving high-risk applicants, which could lead to profit losses or broader financial consequences².

The model is not intended to operate in isolation. Human oversight remains critical—both during deployment and in post-decision review. By positioning the system as a decision-support tool rather than a fully autonomous agent, we preserve flexibility and accountability. Personnel can intervene in cases of borderline scores, anomalies, or client appeals, ensuring that automation complements rather than replaces human judgment.

Interpretability was a guiding principle. Logistic Regression was used not only as a baseline but also to ensure transparency in variable contributions. Decision Trees added an additional layer of explainability by offering a rule-based view of decision boundaries. Even in more complex models like Gradient Boosting, we applied feature importance tools and lift curves to maintain insight into how decisions are made. Performance metrics were evaluated not only globally but also across applicant subgroups (e.g., income quartiles, education levels), ensuring that the model performs consistently and fairly.

All variables were handled in accordance with best practices to avoid discriminatory bias. Sensitive attributes—such as age, education level, employment length, and savings—were included only after verifying that they contributed meaningfully to model performance without introducing unfair disparities. The model must remain unbiased and respect the principle of fairness unless specific variables are proven—through statistical and ethical review—to be legitimate predictors of loan repayment behavior.

In addition, assumptions and technical choices were made transparent through detailed documentation. We explicitly reported feature selection criteria, model limitations, and subgroup evaluations. This supports both regulatory compliance and internal alignment with Prometis Bank's risk management framework. The workflow includes procedures for retraining and monitoring the model over time, which will be essential to ensure ongoing compliance as data distributions shift or new risk factors emerge.

Final Recommendation

Based on performance and compliance trade-offs, we recommend the Gradient Boosting model for deployment. It delivered the highest recall and AUC, making it particularly effective at identifying high-risk applicants. However, in contexts where transparency is critical—such as regulatory audits or sensitive client interactions—Logistic Regression remains a strong alternative due to its interpretability and consistency.

To balance these priorities, Prometis Bank could adopt a hybrid strategy: use Gradient Boosting for internal risk scoring and operational efficiency, while relying on Logistic Regression as an explanatory tool for decisions requiring justification. Moving forward, improvements could include integrating rejected applications to reduce selection bias, introducing model monitoring to track performance drift, and establishing a structured human-in-the-loop process to oversee borderline or high-impact cases.

With additional time, the analysis could have been expanded by incorporating time-series elements (e.g., recent account activity), credit bureau scores, or behavioral data such as payment history or transaction patterns. These features could enhance early risk detection and improve robustness. Similarly, testing advanced techniques like SHAP values for local interpretability or fairness metrics across gender and ethnicity would strengthen real-world readiness and institutional accountability.

²Historically, the failure to properly assess credit risk, such as during the 2008 financial crisis, has contributed to the formation of financial bubbles and widespread economic fallout.